

ANTHONY CONDEZO

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OBJECTIVE

Undergraduate computer engineering student with hand-on experience building distributed robotics control systems, concurrent networking software, embedded-style applications, and database optimisation. Seeking opportunities in industrial automation, embedded systems, or distributed infrastructure where I can apply systems-level thinking and low-level programming skills. Open to broader electrical and quantitative engineering roles in power systems or technical finance environments.

PROJECTS | GITHUB: [HTTPS://GITHUB.COM/ANTHONYCONDEZO/TECHNICAL-PROJECTS.GIT](https://github.com/anthonycondezo/technical-projects.git)

ROVER PORTAL – Distributed Networked Robotics Control Platform

ESP32 (Wi-Fi) & ATmega328p (Motor control & RF TX/RX), Embedded C

- Designed a distributed embedded architecture separating high-level TCP command handling (ESP32) from real-time motor control (ATmega328p) via radio control signal.
- Distributed network includes a proxy server; built to forward REACT SPA rover commands to ESP32 (via Wi-Fi) on local LAN. Proxy server served as the TCP server and ESP32 served as the TCP client.
- Implemented interrupt-driven UART communication between microcontrollers (ESP32 & ATmega328p) for deterministic command parsing.
- Built a custom driver for rf-transceiver module NRF24L01 to allow radio tx/rx. SPI communication was used between module and ATmega328p.
- Developed custom command protocol for remote motor control over Wi-Fi.
- Integrated RF and Wi-Fi subsystems with state-based motor logic. Both provided real-time rover control.

READY SET HIRE – AI-powered Interview Platform

React, Node.js, PostgreSQL, PostgREST, Hugging Face Transformers.js, OpenAI API

- Designed and implemented a full-stack interview platform enabling asynchronous AI-assisted job interview.
- Integrated client-side speech-to-text transcription using Hugging Face & Transformers.js (Xenova/whisper-tiny) to run directly in browser.
- Integrated audio capture pipeline using MediaRecorder API with waveform decoding and resampling to 16kHz from Whisper compatibility.
- Designed JWT-based authentication and role-scoped database access via PostgREST.
- Developed SQL schema and initialization scripts enabling self-hosted database and API deployment.
- Integrated OpenAI API (separated Node.js service) for AI-driven interview question generation.

FORMBASE – Dynamic Mobile Data Collection Platform

React Native (Expo_), PostgreSQL, PostgREST, JWT

- Designed and implemented a mobile-first, schema-driven form build enabling dynamic creation of custom data-collection workflows.
- Architected RESTful backend using PostgreSQL & PostgREST with JWT-secured access and database initialisation scripts for self-hosted deployment.
- Built dynamic field rendering engine supported text, dropdown, numeric, image/camera and geolocation inputs.
- Developed filter criteria-builder supporting numeric and string operations with AND/OR logical composition.
- Designed record persistence layer storing form response as structured JSON (including geospatial and media metadata).

SKILLS

PROFESSIONAL SKILLS

- TEAM COLLABORATION
- PROBLEM-SOLVING & ADAPTABLE LEARNER
- MENTORING & LEADERSHIP
- BILINGUAL (ENGLISH/SPANISH)
- //TODO: ADD MORE FROM PREVIOUS RESUME

PROGRAMMING LANGUAGES

- C/C#
- PYTHON
- JAVASCRIPT
- SQL/POSTGRESQL
- HTML
- CSS

FRAMEWORK & LIBRARIES

- REACT & REACT NATIVE
- NODEJS
- EXPRESS
- BOOTSTRAP
- EXPO
- REACT-NATIVE-PAPER

DATABASE SYSTEMS

- SCHEMAS
- TRIGGERS
- STORED PROCEDURES
- QUERY OPTIMISATION

TOOLS & SYSTEMS

- PROCESS MANAGEMENT (LINUX)
- SIGNALS (LINUX)
- TCP NETWORKING (LINUX)
- GIT & SVN
- OPENCV
- REST API
- JSON
- NPM/NPX

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UQFACE – Distributed Facial Recognition System

C, OpenCV, TCP, Multithreading

- Developed a networked client-server system for concurrent facial detection and replacement requests.
- Implemented uqfacedetect, a multi-threaded TCP server capable of handling multiple simultaneous clients, with mutual exclusion to protect shared image data and classifier structures.
- Implemented uqfaceclient, a command-line client that constructs requests, sends images for detection or replacement, and saves server responses.
- Integrated OpenCV Haar cascade classifiers for face and eye detection, enabling image manipulation and replacement.
- Added robust error handling for invalid requests, image errors, and connection failures.
- Implemented connection limiting, SIGHUP-triggered runtime statistics, signal handling, and strict memory management to ensure stable, high-performance operation.

UQPARALLEL – Parallel Task Execution Program

C, Unix, Processes, Signals

- Developed uqparallel, a robust C program to execute multiple tasks concurrently, with flexible input from (1) command line, (2) a file, or (3) stdin.
- Implemented task pipelining; and parallel execution, dynamically managing child processes to maximise concurrency without exceeding a specified limit.
- Implemented signal handling for SIGINT and SIGTERM for clean termination of all child processes and printing appropriate error messages.
- Ensured proper memory management, freeing dynamically allocated memory in parent process and preventing orphaned or zombie processes.
- Managed pipeline execution edge cases, including blocked pipes and combined limits on parallel jobs, preventing deadlocks.

DATABASE SYSTEMS & OPTIMISATION

PostgreSQL

- Designed relational schemas with triggers and stored procedures enforcing business rules.
- Optimized query performance using materialized views and function-based indexes.
- Analyzed execution plans to compare sequential v. index scan strategies.

EXPERIENCE

STEM Ambassador- AJSP | Dec 2025 - Present

Guide a group of Japanese high school students for a one-day STEM immersion, showcasing university facilities and explaining experiences as an Engineering and Computer Science student. Lead campus and Brisbane tutors while providing insights into STEM study pathways and university life.

FRONT-OF-HOUSE STAFF - THE BRISBANE CLUB | SEPT 2022 – JUNE 2024

Managed team coordination (i.e. kitchen and floor staff) and client interactions during formal events (i.e. handling complaints). Developed leadership, problem-solving, and time-management skills in high-pressure settings.

Team Member – Subway | Dec 2021 – Aug 2023

Entrusted with store supervision and duties coordination. Trained staff and ensured operational efficiency. Strengthening attention to detail, fostered teamwork, and reliability in a fast-paced environment.

MATHEMATICS TUTOR - CLUEY LEARNING | AUG 2021 – MAR 2022

Provided personalized tutoring, enhancing students' problem-solving and analytical skills while fostering a growth mindset.

EDUCATION

Bachelor of Engineering (Honors) / Computer Science

University of Queensland, St Lucia | 2022 - Present

Clairvaux Mackillop College

Upper Mt Gravatt, QLD | 2014 – 2019

REFERENCES

Available upon request.